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# Post-Quantum Secure Communication Test Plan Summary for Industrial Robotic Production Cells

Prototype Test Plan Summary

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## 1 Purpose

This document summarises the test plan for the current post-quantum secure industrial telemetry prototype, focusing on the OT–GA interface. The executed tests verify that the Global Aggregator rejects malicious, stale, modified, replayed, or unauthorised telemetry using ML-DSA signature verification, certificate validation, timestamp freshness checks, and counter-based replay protection. GA–IT security testing with ML-KEM, HKDF, and AES-256-GCM is planned as future work.

## 2 Current Test Scope

The current validation focuses on the OT–GA interface. This is the part of the system where untrusted telemetry first reaches the Global Aggregator and where malicious packets must be rejected before they are accepted as trusted data.

| Test Area                      | Interface Tested               | Status      | Main Security Goal                                   |
|--------------------------------|--------------------------------|-------------|------------------------------------------------------|
| OT–GA Security                 | OT device to Global Aggregator | Executed    | Authenticity, integrity, freshness and device trust. |
| GA–IT Secure Communication     | Global Aggregator to IT layer  | Future work | Confidentiality, integrity and session security.     |
| System Robustness and Recovery | Full prototype                 | Future work | Stability, restart recovery and safe failure.        |

Table 1: Current and Future Test Scope

## 3 Executed OT–GA Layer Security Tests

The following six tests were executed to validate whether the Global Aggregator correctly rejects malicious or invalid telemetry before it is accepted as trusted data.

| No. | Attack Scenario        | Attack Method                                                    | Expected Result                                       |
|-----|------------------------|------------------------------------------------------------------|-------------------------------------------------------|
| 1   | Replay Attack          | Resend a previously captured packet with the same counter.       | Packet blocked; replay attack detected.               |
| 2   | Stale Packet Injection | Delay telemetry packet by more than 5 seconds before forwarding. | Packet blocked; stale packet detected.                |
| 3   | Payload Modification   | Modify sensor and actuator values after packet signing.          | Packet rejected; ML-DSA signature verification fails. |

| No. | Attack Scenario      | Attack Method                                          | Expected Result                                  |
|-----|----------------------|--------------------------------------------------------|--------------------------------------------------|
| 4   | Signature Corruption | Replace the packet signature with invalid bytes.       | Packet rejected; invalid ML-DSA signature shown. |
| 5   | Certificate Forgery  | Replace the valid certificate with a fake certificate. | Packet rejected; invalid certificate shown.      |
| 6   | Unauthorized Device  | Send telemetry from an unregistered public key.        | Access denied; device not trusted.               |

Table 2: Executed OT-GA Security Tests

## 4 General Test Execution Procedure

The same general procedure was followed for each executed OT-GA test.

| Step | Action                                                                                   |
|------|------------------------------------------------------------------------------------------|
| 1    | Start the OT device, signing module, Global Aggregator and dashboard.                    |
| 2    | Confirm that valid telemetry is generated, signed and accepted under nominal conditions. |
| 3    | Capture or prepare a valid telemetry packet for replay or modification tests.            |
| 4    | Launch one attack scenario using the attack script.                                      |
| 5    | Observe whether the Global Aggregator accepts, rejects or blocks the packet.             |
| 6    | Check that the dashboard or logs show the expected security result.                      |
| 7    | Record whether the success criteria were satisfied.                                      |

Table 3: General Test Execution Procedure

## 5 Pass/Fail Criteria

A test is considered **passed** if the Global Aggregator produces the expected rejection behaviour without crashing and without allowing malicious, unauthorised, stale or modified telemetry to be accepted as valid.

A test is considered **failed** if any of the following occurs:

- A replayed packet is accepted.
- A stale packet is accepted.

- A modified payload passes ML-DSA signature verification.
- A forged certificate is trusted.
- An unauthorised device is accepted.
- The Global Aggregator or dashboard crashes unexpectedly.

## 6 Future Test Plan: GA–IT Secure Communication

The next stage of testing will validate the secure communication channel between the Global Aggregator and the IT layer. These tests will focus on ML-KEM key establishment, HKDF-based traffic key generation and AES-256-GCM authenticated encryption.

| No. | Planned Test Scenario                 | Objective                                  | Expected Result                               |
|-----|---------------------------------------|--------------------------------------------|-----------------------------------------------|
| 1   | Encrypted Payload Tampering           | Validate AES-GCM integrity protection.     | Decryption or authentication failure.         |
| 2   | Man-in-the-Middle Packet Modification | Validate secure channel tamper resistance. | Modified encrypted telemetry rejected.        |
| 3   | Wrong Kyber Ciphertext                | Validate ML-KEM session security.          | Shared secret recovery fails.                 |
| 4   | Packet Sniffing                       | Validate confidentiality.                  | Captured traffic does not reveal plaintext.   |
| 5   | Traffic Key Rotation Validation       | Validate HKDF key separation.              | Unique traffic keys generated across packets. |
| 6   | Nonce Reuse Attempt                   | Validate AES-GCM nonce safety.             | Integrity anomaly detected.                   |

Table 4: Future GA–IT Secure Communication Tests

## 7 Future Test Plan: Robustness and Recovery

Additional future testing will evaluate whether the prototype remains stable during operational interruptions and higher traffic conditions.

| No. | Planned Test Scenario   | Objective                        | Expected Result                          |
|-----|-------------------------|----------------------------------|------------------------------------------|
| 1   | Network Flooding        | Validate gateway robustness.     | GA remains operational without crashing. |
| 2   | Device Restart Recovery | Validate OT device reconnection. | Secure telemetry resumes after restart.  |

| No. | Planned Test Scenario     | Objective                     | Expected Result                             |
|-----|---------------------------|-------------------------------|---------------------------------------------|
| 3   | GA Layer Restart Recovery | Validate gateway recovery.    | System reinitialises and resumes operation. |
| 4   | IT Layer Restart Recovery | Validate monitoring recovery. | Dashboard resumes displaying telemetry.     |

Table 5: Future Robustness and Recovery Tests

## 8 Limitations

The current validation focuses only on the OT–GA interface. It demonstrates that the Global Aggregator can reject replayed, stale, modified, incorrectly signed and unauthorised OT telemetry.

However, the current testing does not yet fully validate the GA–IT secure channel under adversarial conditions. Future work will include direct tests of ML-KEM session establishment, AES-GCM encrypted payload protection, packet sniffing resistance, traffic key rotation and system recovery under failure conditions.